

Working with the Charité high performance cluster (HPC)



- 1) Obtain access to the **cluster**: write an email to sc-hpc-helpdesk@charite.de requesting access to the cluster from your charité email mentioning your Charité username (typically 5-7 characters, alphanumerical)
 - 2) Obtain access to the **cluster documentation**: visit <https://git.bihealth.org/charite-sc-public/sc-wiki/> , when first trying to access this, you are prompted to apply for access, which you should do (one sentence description of what you want to do with the cluster is sufficient) and which will hopefully be granted quickly
- Access the cluster via command line at
 - `ssh <username>@s-sc-frontend1.charite.de`

Working with the Charité high performance cluster (HPC)



```
██████████@s-sc-frontend2:~ — ssh ██████████@s-sc-frontend2.cha...
Last login: Mon Jun  2 10:42:43 on ttys000
██████████ ~ % ssh ██████████@s-sc-frontend2.charite.de
[██████████@s-sc-frontend2.charite.de's password:
Welcome: Scientific Computing Team @Charite
This is frontend node s-sc-frontend2.charite.de of the Charite
Compute Cluster.

You can find the User Documentation, various HOWTOs and contact
information in our Gitlab Wiki:

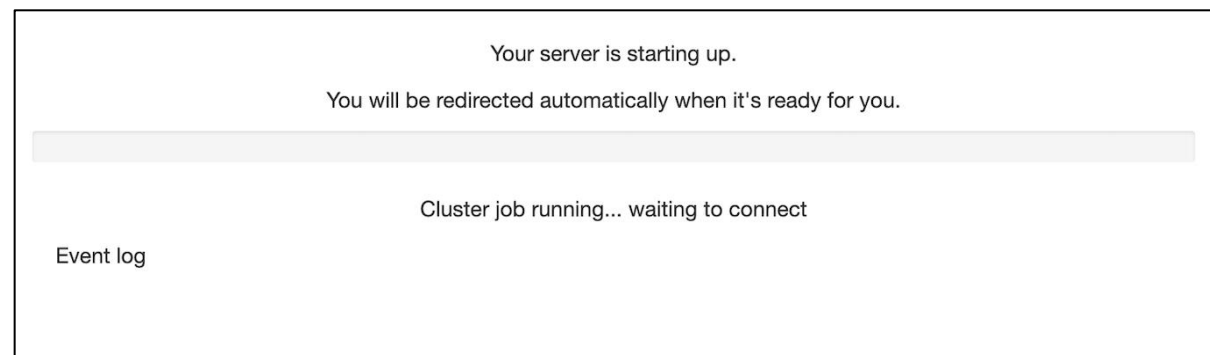
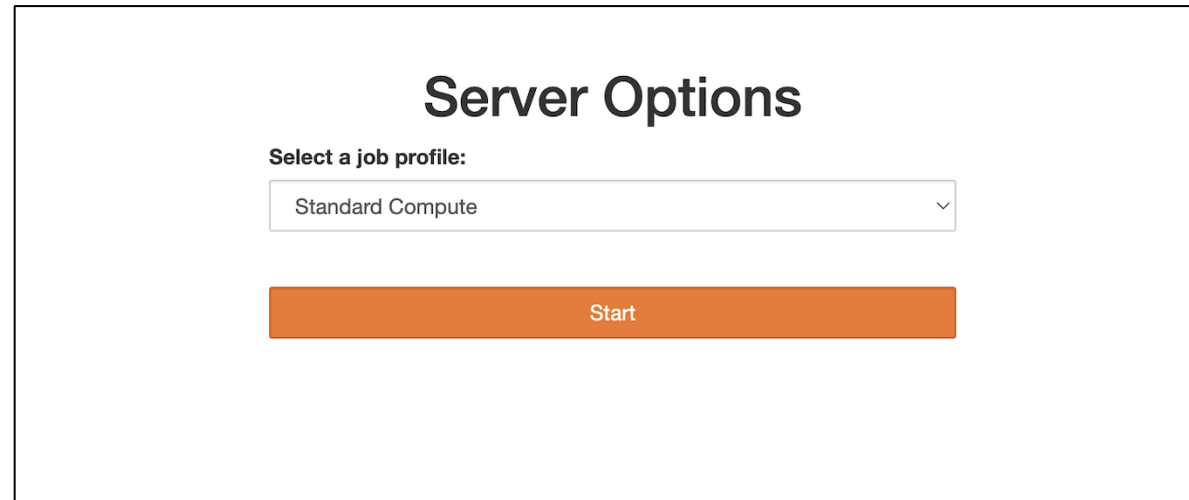
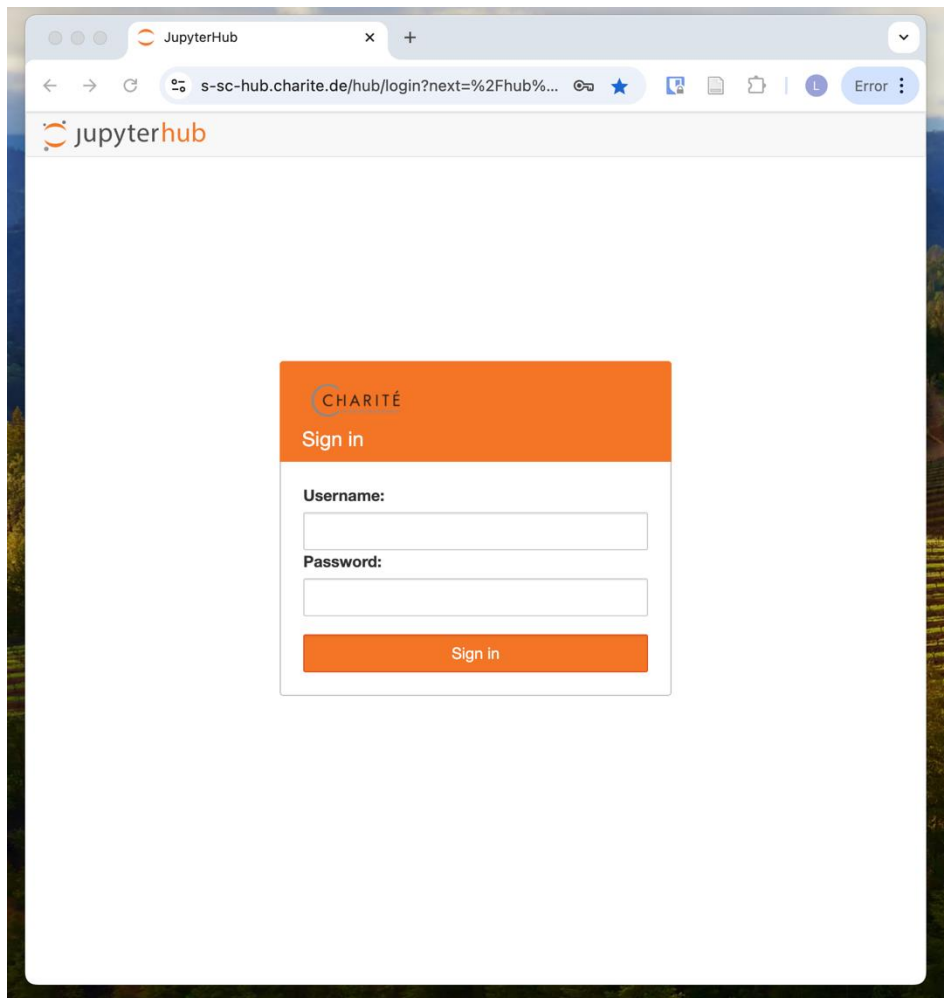
https://git.bihealth.org/charite-sc-public/sc-wiki
Have a lot of fun!: Scientific Computing Team @Charite
Last login: Mon Jun  2 10:42:51 2025 from 10.43.72.188
[██████████@s-sc-frontend2 ~]$ ls
PBMC_lens  ards  covid_lav  data  examples  projects  scratch.txt  software
[██████████@s-sc-frontend2 ~]$
```

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- Access the cluster via command line at
 - `ssh <username>@s-sc-frontend1.charite.de`
 - **Access the jupyter hub** (recommended for beginners and interactive analyses in python) at
 - <https://s-sc-hub.charite.de/hub/>
 - Or **access the on demand portal** (recommended for beginners and interactive Rstudio sessions) at
 - <https://s-sc-ood.charite.de/>

Working with the Charité high performance cluster (HPC) / jupyter hub



Working with the Charité high performance cluster (HPC) / jupyter hub



JupyterLab (auto-G)

s-sc-hub.charite.de/user/.../lab/workspaces/aut...

File Edit View Run Kernel Git Tabs Settings Help

Launcher

Filter files by name

Name Last Modified

- ards 3 months ago
- covid_lav last year
- data 9 months ago
- examples 24 days ago
- PBMC_lens last year
- projects 2 months ago
- software 18 days ago
- jupyterhub... 20 seconds ago
- scratch.txt last year

Other

- Terminal
- Text File
- Markdown File
- Python File
- R File
- Show Contextual Help
- Jupyter
- Python
- R

Simple 0 0 0 Launcher 0

D4_metacells... (auto-G) - Ju x +

s-sc-hub.charite.de/user/.../lab/workspaces/aut...

File Edit View Run Kernel Git Tabs Settings Help

Launcher

Filter files by name

Name Last Modified

- metaout 2 months ago
- A-r_Hughe... 2 months ago
- archive_D... last month
- B-r_Hughe... 2 months ago
- C-p_plotti... last month
- circ_meta... last month
- circ_meta... last month
- circadian_... 2 months ago
- circadian_... 2 months ago
- CircMed_... last month
- D1_merge... last month
- D2_make_... 28 days ago
- D3_circadi... 28 days ago
- D4_metac... last month
- Droin_adat... last month
- Droin_circ... last month
- Droin_mt_... 28 days ago
- GSE11923... 2 months ago
- GSE15913... 4 years ago
- Hughes_2... last month
- Hughes_2... 2 months ago
- Hughes_2... 2 months ago
- Hughes_2... 2 months ago

Metacells object from circadian gene space

```
[40]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scanpy as sc
import anndata as ad
sns.set_context('paper')
import metacells as mc

import utils
import plots
from importlib import reload
reload(utils)
reload(plots)
from utils import sample_colors, sum_counts_by_
from plots import plot_scatter_gene_expression_

from scipy.stats import circmean
```

```
[60]: # read single cell data
adata_circadian = ad.read_h5ad("Droin_circadian_
adata_circadian.obs['time'] = adata_circadian.c
#adata_circadian.var_names_make_unique()
```

```
[61]: adata_circadian.X = adata_circadian.layers["cou
```

```
[62]: adata_circadian.X.sum(axis = 1).mean()
```

```
[62]: 21.241701
```

```
[63]: adata_all = ad.read_h5ad("Droin_mt_ribo_filtere
```

Simple 0 1 0 chrono | Idle Mode: Command Ln 1, Col 1 D4_metacells.ipynb 0



Working with the Charité high performance cluster (HPC) / Rstudio

Sign in to charite

s-sc-sso.charite.de/realms/charite/login-actions/authentic...

CHARITE

Sign in to your account

Username or email

libu10

Password

.....

Sign In

Dashboard - Open OnDemand

s-sc-ood.charite.de/pun/sys/dashboard

Open OnDemand

Files

Jobs

Clusters

Interactive Apps

Desktops

Cluster Desktop

GUIs

MATLAB (legacy)

MEGA

QuPath

Relion

facemap

Servers

Code Server

RStudio Server

OPEN OnDemand

OnDemand provides an integrated, single access point for all of your HPC resources.

powered by OPEN OnDemand

OnDemand version: 4.0.11

https://s-sc-ood.charite.de/pun/sys/dashboard/batch_connect/sys/sc-rstudio/session_contexts/new



Working with the Charité high performance cluster (HPC) / Rstudio

RStudio Server - Open OnDemand

s-sc-ood.charite.de/pun/sys/dashboard/batch_connect/sys/sc-...

RStudio Server

RStudio Server version: 0.0.2

This app will launch an RStudio server on one or more nodes.

Account

sc-users

Partition

agpu

Number of cores

8

The number of cores.

Memory

32

The amount of memory in GB.

Number of GPU shards

0

The number of GPU shards.

Number of hours

1

rstudio image

/sc-software/singularity-images/rstudio/rstudio_latest.sif

select alternative rstudio sif image

Select Path

Launch

* The RStudio Server session data for this session can be accessed under the [data root directory](#).

powered by **OPEN OnDemand**

OnDemand version: 4.0.11

My Interactive Sessions - Open OnDemand

s-sc-ood.charite.de/pun/sys/dashboard/batch_connect/sessions

Open OnDemand

Session was successfully created.

Home / My Interactive Sessions

Interactive Apps

Desktops

- Cluster Desktop

GUIs

- MATLAB (legacy)
- MEGA
- QuPath
- Relion
- facemap

Servers

- Code Server
- RStudio Server

RStudio Server (10687129) 1 node | 8 cores | Starting

Created at: 2026-09-10 10:25:12 CEST

Time Remaining: 59 minutes

Session ID: 4594b14-3edc-4c78-b726-e1145411f209

Your session is currently starting... Please be patient as this process can take a few minutes.

powered by **OPEN OnDemand**

OnDemand version: 4.0.11



Working with the Charité high performance cluster (HPC) / Rstudio

My Interactive Sessions - Open OnDemand

s-sc-ood.charite.de/pun/sys/dashboard/batch_connect/sessions

Open OnDemand

Session was successfully created.

Home / My Interactive Sessions

Interactive Apps

Desktops

- Cluster Desktop

GUIs

- MATLAB (legacy)
- MEGA
- QuPath
- Relion
- facemap

Servers

- Code Server
- RStudio Server

RStudio Server (10687129) 1 node | 8 cores | Running

Host: s-sc-node015.charite.de

Created at: 2026-09-10 10:25:12 CEST

Time Remaining: 57 minutes

Session ID: 45843d14-3edc-4c78-b72e-e114541ff209

Connect to RStudio Server

powered by OPEN OnDemand

OnDemand version: 4.0.11

My Interactive Sessions - Open OnDemand

RStudio Server

s-sc-ood.charite.de/rnode/s-sc-node015.charite.de/54017/

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

libu10

Posit Assistant

Project: (None)

Console Terminal Background Jobs

R 4.5.3

R version 4.5.3 (2026-03-11) -- "Reassured Reassurer"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Error in .rs.downloadFile(url = "https://cdn.posit.co/posit-ai/manifest.json", :
cannot open URL 'https://cdn.posit.co/posit-ai/manifest.js
on'
In addition:
Warning message:
In .rs.downloadFile(url = "https://cdn.posit.co/posit-ai/ma
nifest.json", :
URL 'https://cdn.posit.co/posit-ai/manifest.json': status
was 'Couldn't connect to server'

Environment History Connections Tutorial

R Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentati

Home

Name	Size	Modified
data		
examples		
jupyterhub_slurmspawner_636...	31.9 KB	Sep 2...
jupyterhub_slurmspawner_658...	88.5 KB	Oct 2...
jupyterhub_slurmspawner_658...	89.1 KB	Oct 2...
jupyterhub_slurmspawner_658...	109.3 KB	Oct 2...
jupyterhub_slurmspawner_744...	108.2 KB	Jan 9,
jupyterhub_slurmspawner_766...	29.3 KB	Jan 27
jupyterhub_slurmspawner_766...	30.6 KB	Jan 28
jupyterhub_slurmspawner_808...	32.3 KB	Mar 1
jupyterhub_slurmspawner_829...	29.3 KB	Mar 2
jupyterhub_slurmspawner_829...	27.3 KB	Mar 2
jupyterhub_slurmspawner_888...	30.8 KB	May 1
jupyterhub_slurmspawner_889...	32.1 KB	May 1
jupyterhub_slurmspawner_893...	108 KB	May 1
jupyterhub_slurmspawner_894...	15.9 KB	May 1
jupyterhub_slurmspawner_894...	90.5 KB	May 1
jupyterhub_slurmspawner_896...	29.3 KB	May 2
jupyterhub_slurmspawner_937...	33.5 KB	Jun 11
jupyterhub_slurmspawner_100...	29.8 KB	Jul 17
jupyterhub_slurmspawner_101...	50.3 KB	Jul 30
jupyterhub_slurmspawner_104...	50.3 KB	Aug 2
ondemand		
projects		
R		
scratch.txt	945 B	May 1

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 - 2) Obtain access to the cluster documentation: visit <https://git.bihealth.org/charite-sc-public/sc-wiki/> , when first trying to access this, you are prompted to apply for access, which you should do and which will hopefully be granted quickly
- Access the cluster via command line at
 - `ssh <username>@s-sc-frontend1.charite.de`
 - Access the jupyter hub or on demand portal at
 - <https://s-sc-hub.charite.de/hub/> or <https://s-sc-ood.charite.de/>
 - **The cluster can be accessed only from within the Charité network**
 - **If you want to work remotely, you need VPN access with extra privileges** (“Zusatzantrag O” filled according to the instructions here <https://git.bihealth.org/charite-sc-public/sc-wiki/-/wikis/Resources/User%20Documentation/User%20Guide:%20HPC%20@Charite#connection-to-the-cluster-from-outside-the-charite-network-over-vpn> and screenshot on the next slide)

Working with the Charité high performance cluster (HPC)



Connection to the cluster from outside the Charite network over VPN

If you need to connect to the cluster through VPN, you have to apply for [VPN permissions](#) (Antrag O). The current list of the relevant hosts and ports is here:

- s-sc-frontend1.charite.de, 22/ssh und 443/https
- s-sc-frontend2.charite.de, 22/ssh und 443/https
- s-sc-frontend3.charite.de, 22/ssh und 443/https
- s-sc-hub.charite.de, 443/https
- s-sc-metrics.charite.de, 443/https

To simplify the VPN application process, the VPN-Team of the Charite IT (vpn@charite.de) has defined the group rule `charite-hpc-user-access`. You can simply enter the name of this group in your VPN application (Erweiterte Berechtigung »Spezifische Dienste« -> Zielsystem: Gruppenregel charite-hpc-user-access)

Note on the VPN application VPN permissions are neither necessary nor sufficient for cluster access. A granted VPN application does not imply login access to the frontend nodes. To be granted access to the cluster, please send a request to the SC Helpdesk as explained above.

Working with the Charité HPC – live demo of jupyter hub, documentation and dashboard



- **Access to jupyter hub** <https://s-sc-hub.charite.de/hub/>
- **Access to the on demand portal for Rstudio** <https://s-sc-ood.charite.de/>
- **General user guide** (includes info on how to use prebuilt conda environments to notebook kernels): <https://git.bihealth.org/charite-sc-public/sc-wiki/-/wikis/Resources/User-Documentation/User-Guide:-HPC-@Charite>
- How to submit jobs to SLURM (large and/or repetitive jobs, like alignment jobs, that do not require interaction) <https://git.bihealth.org/charite-sc-public/sc-wiki/-/wikis/Resources/User-Documentation/User-Guide:-HPC-@Charite>
- **Dashboard** of the cluster, shows how busy it is currently <https://s-sc-metrics.charite.de/>
- Applying for larger data storage folders works via email, as outlined here (PI or data owner involvement required) <https://git.bihealth.org/charite-sc-public/sc-wiki/-/wikis/Resources/User-Documentation/User-Guide:-HPC-@Charite#data-storage>
- Connecting **Rstudio** to the cluster is possible, but requires some fiddling <https://git.bihealth.org/charite-sc-public/sc-wiki/-/wikis/Resources/HOWTOs/Rstudio>

Of note - Charité HPC is not the same as BIH HPC



- In addition to the system described above, BIH also provides a Compute Cluster which can be used by Charité and MDC users
- The BIH HPC is extremely well documented, see here <https://hpc-docs.cubi.bihealth.org/>
- Obtaining access works via the group leader who first needs to apply for a user group to be installed
- Similar to Charité HPC, it also includes an interactive portal, the OnDemand portal <https://hpc-portal.cubi.bihealth.org/pun/sys/dashboard>
- This allows interactive jupyter notebook sessions as well as Rstudio sessions